

Drinking water quality assessment through multivariate statistical approach and formulated PCA-based contamination index

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ABSTRACT

Ensuring safe drinking water remains a major public health and sustainability challenge in rapidly urbanizing, groundwater-stressed and semi-arid regions. Conventional monitoring lacks statistical integration and holistic assessment. This study evaluated 167 drinking water samples from surface (PHED supply from Bisalpur Dam) and groundwater sources, assessing parameters including physicochemical, heavy metals, bacteriological indicators and antibiotic resistance profiles of 141 *E. coli*-like isolates. To enhance interpretability and risk stratification, an advanced multivariate statistical framework was applied to derive a novel Principal Component Analysis-based Drinking Water Contamination Index (DWCI), supported by Mahalanobis Distance (MD) outlier detection and K-means clustering with ANOVA-derived F-values for variable importance. Key findings revealed significant contamination: multiple samples exceeded WHO limits for TDS, fluoride, lead and chromium. Microbial analysis showed high microbial load (3.00–6.44 log₁₀ CFU/mL), with 36.53% of samples contaminated by *E. coli* and total coliforms (1.8–1800 MPN/100 mL) frequently surpassing safe thresholds, highlighting fecal contamination and infrastructural vulnerabilities. 93.6% of isolates from *E. coli*-positive samples ($n = 61$) exhibited resistance to multiple antibiotics/drugs, with 96.45% surpassing high-risk multiple antibiotic resistance index (MARI) threshold. Groundwater consistently exhibited higher contamination across all assessed parameters. MD analysis identified 22 outlier samples (13 surface, 9 groundwater) with extreme contamination, overlapping with high DWCI values, while clustering revealed three contamination regimes, from baseline geogenic to severe anthropogenic pollution. This study demonstrates the effectiveness of integrated multivariate techniques in characterizing contamination patterns, offering a replicable, cost-effective framework for water quality monitoring, risk-based regulation, supporting SDG-6 targets for safe, equitable drinking water access in water-stressed environments.

1. Introduction

The global drinking-water crisis remains a critical and multifaceted challenge, with approximately 4.4 billion people across 135 low- and middle-income countries lacking access to safe drinking water – a sharp rise from earlier estimates of over 2 billion in 2020 (Hope, 2024). This escalating insecurity is driven by a confluence of factors: weak water governance, inadequate institutional capacity, and the over-extraction and pollution of freshwater, which have already degraded >60% of ecosystem services (Kammeyer et al., 2020). Simultaneously, accelerating urbanization, demographic expansion, and climate change, are expected to increase global water demand by nearly 40% by 2030 (Guyman, 2024), threatening progress toward Sustainable Development Goal 6 (SDG-6), which recognizes access to clean water as a basic human

right (Eman and Meško, 2020).

Beyond scarcity, deteriorating water quality is an equally pressing concern arising from both anthropogenic and natural sources. Anthropogenic pressures are dominant; over 80% of global wastewater is discharged untreated, contaminating surface and groundwater (Kammeyer et al., 2020), while rapid industrialization, unregulated agricultural runoff, and urban encroachment exacerbate microbial and heavy metal (HM) contamination (Gupta and Chinnasamy, 2022). Naturally occurring pollutants such as fluoride, arsenic, and nitrate further impair groundwater, causing chronic health conditions (Ahmad et al., 2023; Panahi et al., 2021). Together, these hazards result in a significant health burden: in 2019, unsafe water, sanitation, and hygiene (WASH) contributed to an estimated 1.4 million deaths (OECD, 2023). These risks are disproportionately severe in low-resource settings such as sub-

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Saharan Africa and South Asia, where inadequate infrastructure and reliance on contaminated water sources perpetuate outbreaks of waterborne diseases and long-term health risks from HM bioaccumulation (Krishan et al., 2023a, 2023b; Zeng et al., 2024).

While these global patterns are concerning, their impacts are particularly severe in South and Southeast Asia, especially India, where rivers receiving untreated sewage and effluents have become reservoirs of antibiotic-resistant bacteria (ARBs) and resistance genes (ARGs), with abundances reaching up to 10^5 copies mL^{-1} (Bagra et al., 2024; Naznine et al., 2024). This challenge is compounded in semi-arid urban areas, where acute water security threats from erratic rainfall and over-exploited groundwater exacerbate chemical, microbial, and antibiotic resistance contamination, posing a formidable monitoring challenge (Analy and Laftouhi, 2021).

Effectively monitoring these diverse water pollutants is fraught with difficulty, as water quality is shaped by spatiotemporal variability, interacting contaminants and complex datasets (Blair et al., 2025). Conventional frameworks often assess physicochemical, microbial, and antimicrobial resistance (AMR) parameters in isolation, resulting in fragmented evaluations. Traditional Water Quality Indices (WQIs), for instance, provide useful initial assessments but are fundamentally limited by their primary reliance on physicochemical parameters, thereby overlooking critical threats from HMs, pathogens, and multidrug-resistant (MDR) bacteria (Abbasi and Abbasi, 2011; Banda and Kumarasamy, 2020; Dwivedi and Dubey, 2023). Even newer machine learning-derived Integrated WQIs (IWQI), have not been systematically extended to incorporate this crucial microbiological and AMR data, leaving a substantial gap in holistic risk assessment (Essamlali et al., 2024; Omeke et al., 2024).

While advanced multivariate statistics show promise, their application in water quality assessment remains siloed. Chemometric analyses, for example, effectively use Principal Component Analysis (PCA) to identify geochemical signatures and pollution sources (Omeke et al., 2024; Omeke and Igwe, 2023), but these studies rarely couple their outputs with microbiological data to construct comprehensive contamination indices (Fu et al., 2020; Mohamed Amine et al., 2021). This disciplinary fragmentation means that powerful methods, including PCA for dimensionality reduction, MD for multivariate outlier detection, and K-means clustering (KMC) for identifying contaminant influence – remain underutilized in an integrated capacity (Mounir et al., 2024; Ngoc Nguyen et al., 2025; UN-Water, 2021). Consequently, no universally accepted, statistically rigorous DWCI currently exists that can synthesize physicochemical, microbial, and AMR datasets within a single, unified framework.

To address these critical gaps, the present study distinguishes itself from existing studies by introducing a novel, cost-effective framework that integrates disparate datasets into a PCA-driven DWCI, augmented with MD and KMC analyses. Unlike previous siloed approaches, this work systematically unifies these statistical methods to create a single, adaptable tool for evidence-based decision-making. This integrated approach enhances contamination ranking, strengthens anomaly detection, and provides regulators with a practical method for prioritizing interventions in diverse hydro-climatic contexts.

Accordingly, the study aims to: (i) assess 21 physicochemical, HM and microbiological parameters in surface and groundwater sources across a representative semi-arid urban setting, to capture spatial heterogeneity; (ii) conduct phenotypic antibiotic susceptibility testing of *E. coli*-like isolates, determine multidrug resistance (MDR) prevalence and compute the Multiple Antibiotic Resistance Index (MARI) as an indicator of environmental antibiotic pressure; (iii) formulate a DWCI using PCA for objective contamination ranking; (iv) identify the most influential contaminants through F-value importance scores obtained from KMC integrated with one-way ANOVA, providing regulators with data-driven prioritization insights; and (v) implement MD analysis for flagging samples that deviate significantly from the multivariate centroid, thus identifying high-risk outliers potentially overlooked by

conventional threshold-based assessments. Together, these objectives operationalize a scalable, data-driven approach that moves beyond fragmented water quality monitoring, providing a robust evidence base for achieving SDG-6 targets in water-stressed environments (Ngoc Nguyen et al., 2025; UN-Water, 2021).

2. Methodology

2.1. Study area

The study was conducted in Jaipur (Fig. 1), Rajasthan, as it exemplifies critical conditions targeted in this investigation: (i) semi-arid climate with hydro-climatic stress and aging infrastructure; (ii) Low annual rainfall (<65 cm) and severe groundwater depletion (>3 m/year), requiring reliance on both distant surface water (Bisalpur Dam – 120 km located) and local groundwater sources; (iii) chronic water quality challenges including fluoride, nitrate, and Total Dissolved Solids (TDS) exceedances in $\sim 70\%$ of the piped supply; and (iv) recurring *E. coli* and AMR contamination. As surface water is supplied from Bisalpur Dam, its chemistry may also reflect the geology of the Bisalpur catchment in addition to local distribution influences, and is therefore considered only as part of the supplied water system. These combined factors make Jaipur an ideal setting for integrated water quality assessment (Khandelwal, 2022; Mangal et al., 2023; Rukshar et al., 2023). The expanded sampling density and analytical scope in this study provide both temporal continuity and complexity, offering insights applicable to other water-stressed regions.

2.2. Sample collection

A total of 167 drinking water samples (51 from borewell, and 116 from Public Health Engineering Department, PHED supply) were collected periodically from diverse locations across Jaipur, Rajasthan, between October 2021 and August 2024 (Fig. 1). This extended sampling period was designed to capture a range of hydro-climatic conditions across multiple seasons, providing a comprehensive long-term assessment of baseline water quality. The samples represented two primary sources of drinking water: i) treated surface water supplied through the PHED distribution network sourced from Bisalpur Dam, and ii) local borewell (groundwater) sources, collected from drinking water outlets (consumer-end points) within the city.

The spatial distribution of sampling sites was guided by a stratified methodology previously described by Srivastava et al. (2025), which combines systematic, random, and targeted approaches to ensure robust socio-economic and geographical representation. Samples were systematically collected at intervals of approximately 2–3 km per ward. This framework included residential, healthcare, commercial, industrial, and slum areas, covering both high- and low-risk zones based on known contamination patterns and infrastructure variability.

Sampling followed APHA (2017) guidelines (Baird et al., 2017), with collection in pre-sterilized bottles that were rinsed with source water to maintain aseptic conditions and preserve the native microbial composition. Samples were transported to the laboratory in an ice-box container at $4\text{--}8^\circ\text{C}$ and processed immediately so that the elapsed time between the sample collection and initial processing did not exceed 24 h. Field quality control measures were ensured included the collection of duplicate samples and field blanks (one of each per 20 samples) to monitor for cross-contamination and procedural consistency.

2.3. Physicochemical and elemental analysis

Physicochemical parameters, including pH, TDS, calcium, chloride, sulphate, nitrate, fluoride, total hardness, and alkalinity, were assessed in all drinking water samples following standard APHA protocols, widely applied in similar water quality studies (Baird et al., 2017; Mesfin et al., 2024; Nadeem et al., 2025; Srivastava et al., 2025). For elemental

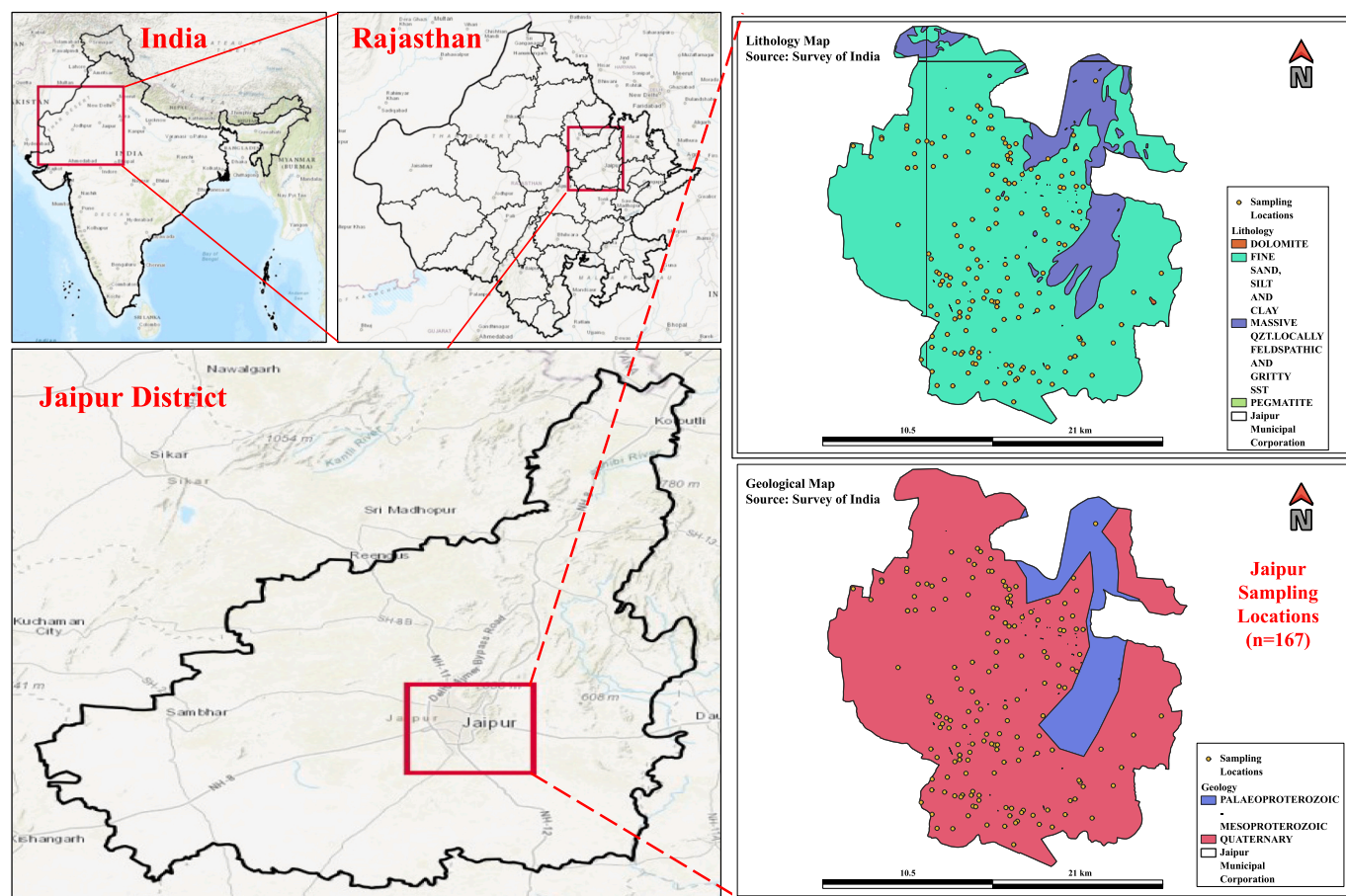


Fig. 1. Geological map of the study area (Jaipur District, Rajasthan, India) showing lithological variations and sampling locations.

analysis, 100 mL acid-preserved samples were digested with 5 mL concentrated nitric acid (HNO_3) at $\sim 95^\circ\text{C}$ on a hot plate until reduced to $\sim 10\text{--}20\text{ mL}$ and a clear digest was obtained, then filtered and diluted to 100 mL with deionized water, following APHA Method 3030E. Concentrations of zinc (Zn), iron (Fe), copper (Cu), Pb, Cr, manganese (Mn), Ni, and phosphorus (P) were quantified using Atomic Absorption Spectrophotometry (AAS) and Inductively Coupled Plasma-Atomic Emission Spectroscopy (ICP-AES) both of which are established techniques for sensitive and reproducible quantification in environmental samples (Lata and Ansari, 2024; Tawfik et al., 2024). Laboratory quality assurance included the use of procedural blanks, calibration with certified standard solutions, and triplicate measurements for each sample to ensure accuracy and reproducibility.

Inductively Coupled Plasma-Atomic Emission Spectroscopy (ICP-AES) is a powerful analytical tool which offers superior sensitivity to measure the trace elements in the environmental samples. The employed techniques provide an optimal balance of accuracy, cost-effectiveness, accessibility and feasibility for routine or large-scale monitoring programs. Therefore, the combination of APHA protocols with AAS and ICP-AES was selected as the most suitable framework, ensuring analytical reliability, in comparison with a vast area of prior research and potential for sustainable, long-term monitoring in the study region.

2.4. Microbiological analysis

Microbiological assessments were conducted following standard protocols (Baird et al., 2017; Srivastava et al., 2025). Total aerobic heterotrophs were determined using the 9215 spread-plate method, while total and thermotolerant coliforms (*E. coli*), were determined by

the 9221 multiple-tube fermentation technique with EMB agar confirmation. A total of 141 *E. coli*-like isolates (77 from surface and 64 from groundwater) were isolated from 167 drinking water samples.

2.5. Antibiotic susceptibility test

Phenotypic antibiotic resistance profiling of 141 *E. coli*-like isolates was conducted using the Kirby-Bauer disk diffusion method, following CLSI guidelines (Weinstein and Lewis, 2020; Srivastava et al., 2025). A total of 20 antibiotics were tested, including amoxicillin (AMX^{30}), ampicillin (AMP^{25}), cefixime (CFM^5), chloramphenicol (C^{30}), ciprofloxacin (CIP^{30}), cloxacillin (COX^{30}), co-trimoxazole (COT^{25}), doxycycline hydrochloride (DO^{30}), erythromycin (E^{15}), gentamicin (GEN^{10}), imipenem (IPM^{30}), kanamycin (K^{30}), methicillin (MET^{30}), nalidixic acid (NA^{30}), neomycin (N^{30}), nitrofurantoin (NIT^{300}), oxacillin (OX^5), polymyxin-B (PB^{300}), rifampicin (RIF^{30}) and tetracycline (TE^{30}). Results were interpreted according to CLSI inhibitory zone criteria. The MARI was calculated for each isolate, with values >0.2 considered high-risk (Asikiya et al., 2023). *E. coli* ATCC 25922 was used as the quality control strain.

2.6. Multivariate statistical analysis

2.6.1. Data preprocessing and standardization

All physicochemical, microbial, and antibiotic resistance data were standardized using Z-score normalization to ensure comparability across variables with differing units and scales, thereby preventing parameters with large numerical ranges (e.g., TDS) from disproportionately influencing the analysis over those with smaller ranges (e.g., HMs). This approach transformed each variable to have a mean of zero and a

standard deviation of one, preparing the dataset for multivariate analyses such as PCA, clustering, and anomaly detection (Hervé, 2022; Mohammed et al., 2022). Non-detectable values were substituted with 0.001 mg/L to address left-censored data, following established practices in environmental statistics (Mohammed et al., 2022). All statistical procedures were conducted using OriginPro software.

2.6.2. Principal component analysis

PCA was then conducted on the standardized data matrix with a dual objective. The primary purpose was dimensionality reduction: to transform the large set of correlated variables into a smaller set of uncorrelated variables, known as principal components (PCs), that capture the maximum variance in the data. This simplifies the complex dataset while retaining essential information. Secondly, the analysis was used for source apportionment: to identify the underlying drivers and potential sources of contamination by examining the loading of different variables onto each PC. Components with eigenvalues >1 and cumulative variance ≥70% were retained following Kaiser's criterion. Variables with absolute loadings ≥0.50 were considered significant contributors (Hamed, 2020; Vidya and Prasad, 2024; Zhao et al., 2020). To enhance interpretability of the component structure, the retained loading matrix was further subjected to Varimax rotation (orthogonal rotation), which redistributes variance among factors without changing the total explained variance (Seth et al., 2025).

2.6.3. Formulation of the drinking water contamination index (DWCI)

A DWCI was formulated for each sample using the Euclidean norm of its standardized unrotated PC scores, derived from the first k principal components (eigenvalues >1) in section 2.6.2, effectively representing the multivariate distance of each sample from the dataset centroid, with higher values indicating greater cumulative contamination. Mathematically, $DWCI_i$ for the i -th sample was computed as:

$$DWCI_i = \sqrt{(PC_{i1})^2 + (PC_{i2})^2 + \dots + (PC_{ik})^2}$$

Where,

$DWCI_i$ = Contamination Index of the i -th sample.

PC_{ij} = Score of the i -th sample along the j -th principal component.

k = Number of retained principal components ($k = 6$).

In the formulation of the DWCI, the use of unrotated PCA scores is critical for maintaining the orthogonality of PCs, which ensures that each component contributes independently to the overall index without overlapping variance. While factor rotation can enhance interpretability by redistributing variable loadings, it often introduces inter-component correlations that can bias quantitative weighting and compromise the mathematical integrity of the index (Krishan et al., 2023a, 2023b). Therefore, unrotated components were used in the DWCI formulation to maintain statistical rigor, prevent redundancy, and ensure comparability with established PCA-based environmental indices (Jewsbury and Johnson, 2025; Mahanty et al., 2023).

2.6.4. Source-wise DWCI boxplot visualization and non-parametric significance testing

Boxplots were generated to visualize differences in DWCI values between surface and groundwater sources. The Mann-Whitney U test was applied to assess statistical significance between these non-normally distributed groups. The p -value was annotated on the plot to support interpretation (Mabude et al., 2022).

2.7. Cluster analysis

2.7.1. K-means clustering (KMC) and variable importance analysis

KMC was performed with $k = 3$, determined using the elbow method and domain knowledge, to classify samples along a contamination gradient. Post-clustering, one-way ANOVA was applied to identify

variables significantly influencing cluster separation, with high F -values and $P \leq 0.05$ indicating key contributors. A standardized cluster-centroid profile (baseline $z = 0$) was constructed to visualize deviations from the mean across variables, aiding the interpretation of distinct water-quality patterns (Hasibuan et al., 2023; Puri and Gupta, 2024).

2.8. Mahalanobis distance (MD) analysis

MD was calculated on the Z-score standardized data matrix (167×21) using the multivariate mean vector and sample covariance matrix, accounting for inter-variable correlations to identify multivariate outlier indicative of extreme contamination events (Cabana et al., 2021; Suboh et al., 2023). The statistical threshold for outlier detection was set at the 99.5th percentile of the chi-square distribution ($\chi^2_{0.995, df = 21}$), yielding a critical χ^2 value of 41.40 and an MD threshold of $\sqrt{41.40} \approx 6.43$. Samples exceeding this threshold were flagged as significant outliers with 99.5% confidence (Bulut, 2020; Teves, 2023).

3. Results and discussion

3.1. Physicochemical and elemental contamination in drinking water samples

3.1.1. Physicochemical profile of drinking water

Descriptive statistics for the measured physicochemical parameters are provided in Supplementary Table 1. This section highlights collective trends, underlying drivers and environmental implications, supported by boxplot analyses (Fig. 2a–i). Based on hydrogeochemical behavior and regional influences, the analyzed parameters revealed three dominant patterns across the study area (Fig. 3): (i) Parameters such as pH, sulphate, and chloride exhibited conservative and stable behavior with limited spatial variability and are governed largely by lithological composition, representing the baseline geochemical (geogenic) signature of the region, (ii) calcium, total hardness, total alkalinity, and fluoride showed greater spatial and temporal variability, reflecting dynamic geogenic processes driven by mineral dissolution and water–rock interactions, and (iii) conversely, elevated concentrations of TDS and nitrate, particularly in surface water sources, indicated anthropogenic inputs, primarily associated with human activities such as agriculture and domestic effluent discharge. This interpretive framework delineating baseline, dynamic geogenic, and anthropogenic controls aligns with established hydrochemical classification schemes (Hota et al., 2023; Goutham et al., 2025; Wali et al., 2023), providing a clear conceptual basis for subsequent detailed analyses.

Parameters such as pH, sulphate, chloride and nitrate displayed stable distributions across surface and groundwater. The narrow inter-quartile ranges and minimal outliers in these parameters indicate strong geogenic control, particularly from carbonate- and sulphate-bearing formations. Groundwater pH remained neutral to mildly alkaline, reflecting carbonate buffering, a characteristic of semi-arid terrains (Adimalla, 2022; Tetali et al., 2024). Chloride and sulphate levels, though slightly elevated in groundwater, remained within WHO limits, suggesting minimal anthropogenic influence, consistent with findings in Ethiopian and Indian aquifers (Kumar et al., 2023; Mengstie et al., 2023; Sharma and Kumar, 2020). Nitrate levels were uniformly low, although surface waters showed occasional outliers, likely due to localized agricultural runoff, consistent with patterns reported in similar hydrogeological settings (Chander et al., 2023; Gan et al., 2022; Jia and Qian, 2025). In contrast, calcium, total hardness, total alkalinity, and fluoride exhibited pronounced differences between surface and groundwater, with consistently higher median concentrations in groundwater, reflecting prolonged water-rock interactions, including carbonate dissolution, gypsum weathering, and fluoride enrichment (Karobi et al., 2022; Kom et al., 2023; Spahić et al., 2023). Boxplot analyses highlighted surface water outliers near carbonate- or fluoride-rich

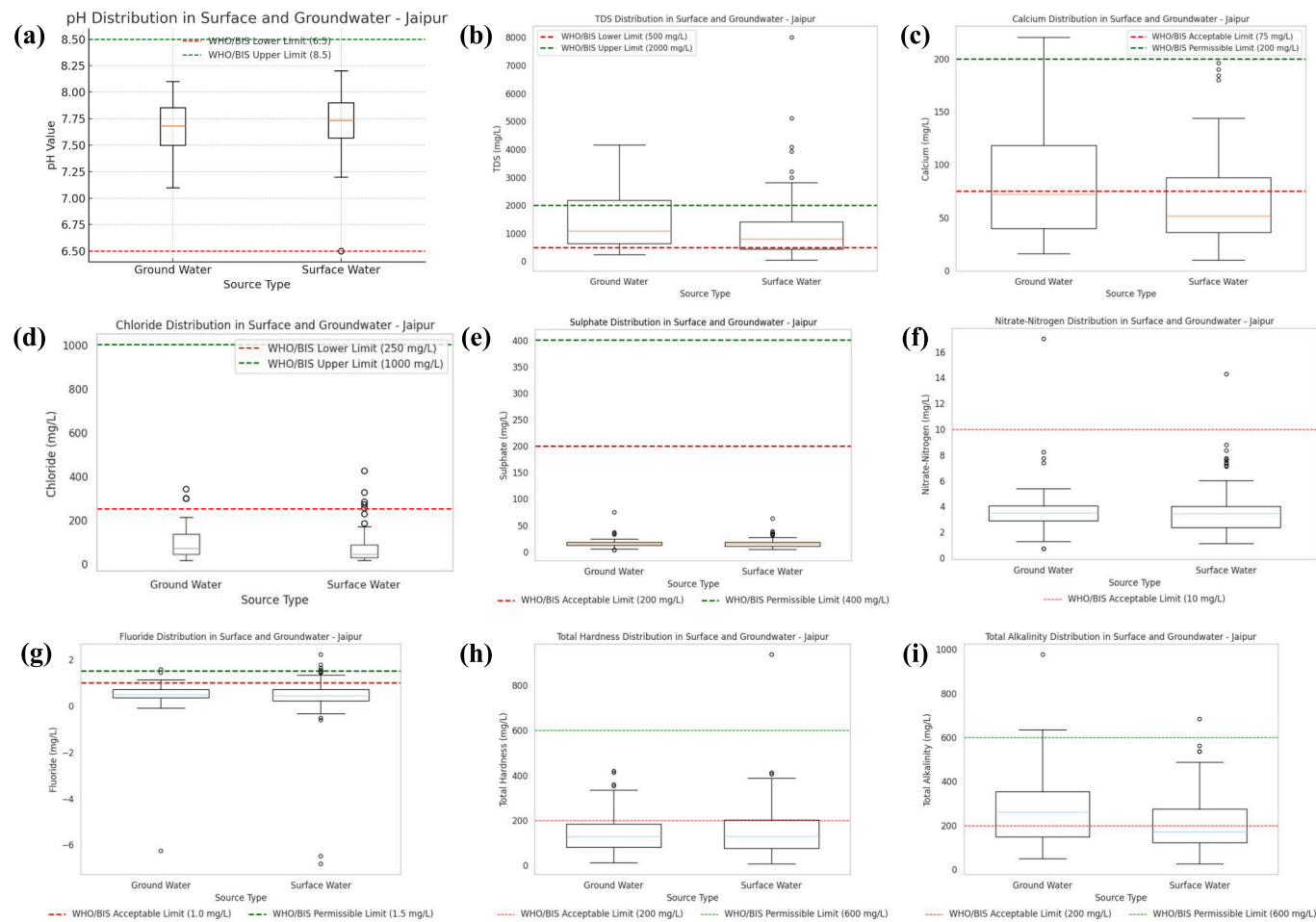


Fig. 2. Box plots showing physicochemical parameters distribution in surface and groundwater samples.

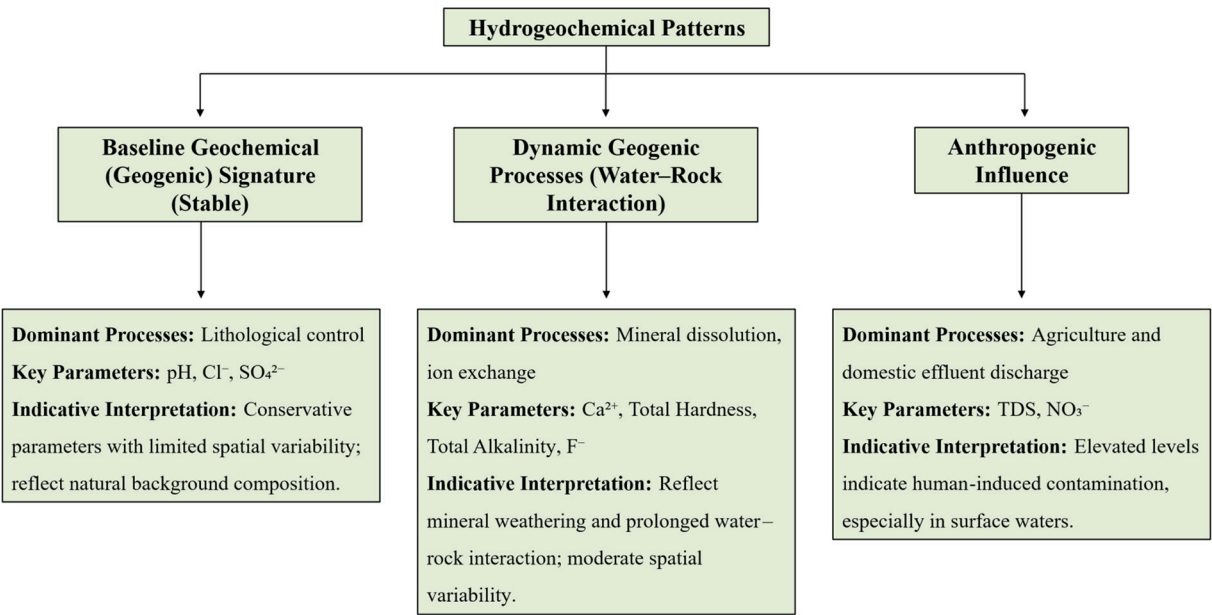


Fig. 3. Conceptual framework delineating hydrogeochemical patterns in drinking water sources across the study area.

lithologies, while groundwater generally remained geochemically stable except for occasional extremes, such as elevated alkalinity in J20, fluoride in J109, J116, and hardness in J15, a pattern consistent with

semi-arid aquifers (Karangoda and Nanayakkara, 2023; Oberhelman et al., 2024; Yasaswini et al., 2024). TDS and nitrate showed the clearest anthropogenic signals; their sources are often intertwined. Elevated TDS

in groundwater, for instance, reflects not only long-term mineral accumulation and salinization but also significantly exacerbated by anthropogenic pressures like the return flow from irrigated agriculture, a process that also elevates nitrate. Similarly, local spikes in both surface and groundwater are linked to effluent discharges, post-monsoon runoff, and widespread contamination from agricultural fertilizers and domestic wastewater, which are key contributors to high nitrate concentrations in water resources (Choudhary et al., 2024; Das et al., 2023; Subba Rao et al., 2024). Boxplots revealed greater surface water variability, particularly in agricultural areas, underscoring the combined impact of geogenic and human-induced pressures (Gan et al., 2022; Jia and Qian, 2025).

Collectively, these findings indicate that while baseline water chemistry is predominantly shaped by natural hydrogeochemical processes, localized anthropogenic pressures, especially on surface water systems, can lead to episodic exceedances. Although most parameters met WHO acceptable or permissible limits, certain hotspots, notably for TDS, alkalinity, and fluoride require targeted management. Persistent exceedances may compromise water palatability, domestic usability, and long-term health, necessitating interventions such as source protection, regular monitoring, and where necessary, treatment interventions such as softening or defluoridation (Giani et al., 2023; Karunanidhi et al., 2022).

3.1.2. Elemental profile of drinking water

Descriptive statistics for measured elements concentration are summarized in Supplementary Table 2. Likewise, key water quality parameters, similar three dominant patterns were observed across the study area for elements (Fig. 4a–i): (i) metals and non-metal largely within acceptable limits reflecting natural geogenic sources (e.g., Zn, Mn, Ni, P); (ii) metals elevated due to lithological enrichment or redox-driven mobilization (e.g., Fe) and (iii) metals with clear anthropogenic signatures showing localized exceedances (e.g., Cu, Pb, Cr).

Zn, Mn, Ni, and P generally showed low median concentrations with occasional surface water outliers. Zn and Ni variability were attributed to mineral leaching and anthropogenic inputs such as urban runoff, pipe corrosion, industrial discharge etc., while P spikes reflected episodic inputs from sewage or fertilizer runoff. Mn mobilization, particularly under reducing conditions, was consistent with patterns reported in arid to semi-arid regions, where Mn is released from Mn-rich minerals or under oxygen-depleted aquifer conditions (Balaram et al., 2023; Friedman et al., 2024; Genchi et al., 2020; Haque, 2021). Fe exhibited elevated concentrations in surface water likely reflecting runoff and weathering due to natural processing as evidenced by Dashdondog et al. (2022) while groundwater showed a lower median but broader variability, potentially resulting from reductive dissolution or lithological enrichment (Abubakar et al., 2023). In contrast, Cu, Pb, and Cr showed the clearest anthropogenic signals. Cu outliers in both surface and groundwater were linked to plumbing corrosion and agrochemical runoff (Manne et al., 2022), while Pb exceedances, notably in surface waters, reflected aging infrastructure and industrial discharge (Dotaniya et al., 2020; Mandal et al., 2023). Cr elevations, associated with industrial inputs like chrome plating and tannery waste, consistent with documented Cr presence in water sources (Gezahegn et al., 2021; Sharma et al., 2022; Younas et al., 2023), raise particular concern due to the carcinogenicity of Cr(VI) (Georgaki et al., 2023). Collectively, while most HMs were within safe limits, episodic exceedances of Fe, Pb, Cr, and Cu reveal the combined influence of natural and human-driven processes on water quality. These hotspots warrant focused monitoring and remediation to ensure long-term drinking water safety (Karim et al., 2020).

3.2. Microbiological characteristics of drinking water

3.2.1. Total aerobic heterotrophic bacterial count

Total aerobic heterotrophic counts ranged from 3.00 to 6.44

log₁₀CFU/mL (mean: 4.70 ± 0.88 ; median: 4.93), with surface water showing greater variability (IQR: 1.59) than groundwater (IQR: 1.07) (Fig. 5). This pattern aligns with previous studies reporting higher microbial loads in surface waters reflecting its greater susceptibility to contamination from anthropogenic sources, whereas groundwater typically exhibits lower and more stable counts, benefiting from natural subsurface filtration and less exposure to surface pollutants (Abdulkareem, 2023; Dong et al., 2024). The absence of statistical outliers in either source suggests a uniform but persistent microbial presence, underscoring the need for continuous monitoring to safeguard public health from waterborne pathogens (Salvador et al., 2020).

3.2.2. Total coliform contamination

The presence of total coliforms in the collected water samples, ranging from 1.8 to 1800 MPN/100 mL (mean: 187 ± 452 ; median: 4), indicates widespread contamination and non-compliance with WHO (2017) standards that mandates the absence of *E. coli* in any 100 mL of drinking water. Boxplot analysis (Fig. 6) showed a surface water median of 2.0 MPN/100 mL (IQR: 44.7) with 21 outliers, indicating episodic fecal contamination from runoff and waste discharge (Some et al., 2021). Groundwater displayed a higher median (6.1) and broader IQR (313), with seven extreme outliers, suggesting infiltration of surface contaminants, a trend noted in compromised well integrity (Sruthi et al., 2022). While not always pathogenic, coliforms signal elevated microbial risk and potential gastrointestinal illness, reinforcing the need for effective disinfection and infrastructure improvements to protect public health (Banseka and Tume, 2024).

3.2.3. *E. coli* contamination

The detection of *E. coli* in the collected water samples, reported at 36.53% (61 out of 167 samples), significantly exceeds the WHO's zero-tolerance guideline, with higher contamination in groundwater (47.06%; 24 out of 51 groundwater samples) than surface water (31.90%; 37 out of 116 surface water samples). Notably, unprotected wells showed contamination rates as high as 71%, indicating very high risk, consistent with previous reports linking *E. coli* presence to inadequate sanitation, proximity to septic systems, and poor source protection. These findings underscore the urgent need for improved water treatment and infrastructure to mitigate health risks (Zgambo et al., 2023).

3.3. Antibiotic resistance profile of *E. coli*-like isolates

The antibiotic resistance among 141 *E. coli*-like isolates (Fig. 7) ranged from 1.4% to 93.6% (mean: $41.9 \pm 24.5\%$), while intermediate resistance ranged from 3.5% to 40.4% (mean: $23.0 \pm 9.9\%$). This indicated strong β -lactam resistance likely driven by environmental factors such as wastewater treatment processes that promote co-evolution of resistance mechanisms (Kasanga et al., 2024). Notably, susceptibility rates varied widely (2.8% for AMP²⁵ to 89.4% for CIP³⁰, mean: 35.04 ± 23.84), with ciprofloxacin demonstrating the highest efficacy, consistent with reports highlighting its retained effectiveness despite global resistance trends (Shariati et al., 2022). These findings underscore the urgent need for robust antibiotic stewardship and systematic environmental AMR surveillance.

The alarming prevalence of MDR *E. coli* in drinking water sources, with 93.62% (132 out of 141) of isolates (originating from 36.53% of all *E. coli* positive water samples) exhibiting resistance to >3 antibiotic classes, underscores a significant public health concern. Studies have reported about 49.48% MDR prevalence among *E. coli* isolates from water sources, with resistance commonly observed against frequently used antibiotics (Kichana et al., 2022; Odonkor and Addo, 2018). In the present study, 59.1% groundwater samples contained MDR bacteria, indicating aquifers as critical reservoirs for MDR pathogens (Kanmani et al., 2023). Supporting evidence from Morocco reported that 58.1% of *Salmonella* spp. isolated from groundwater were MDR (Hafiane et al.,

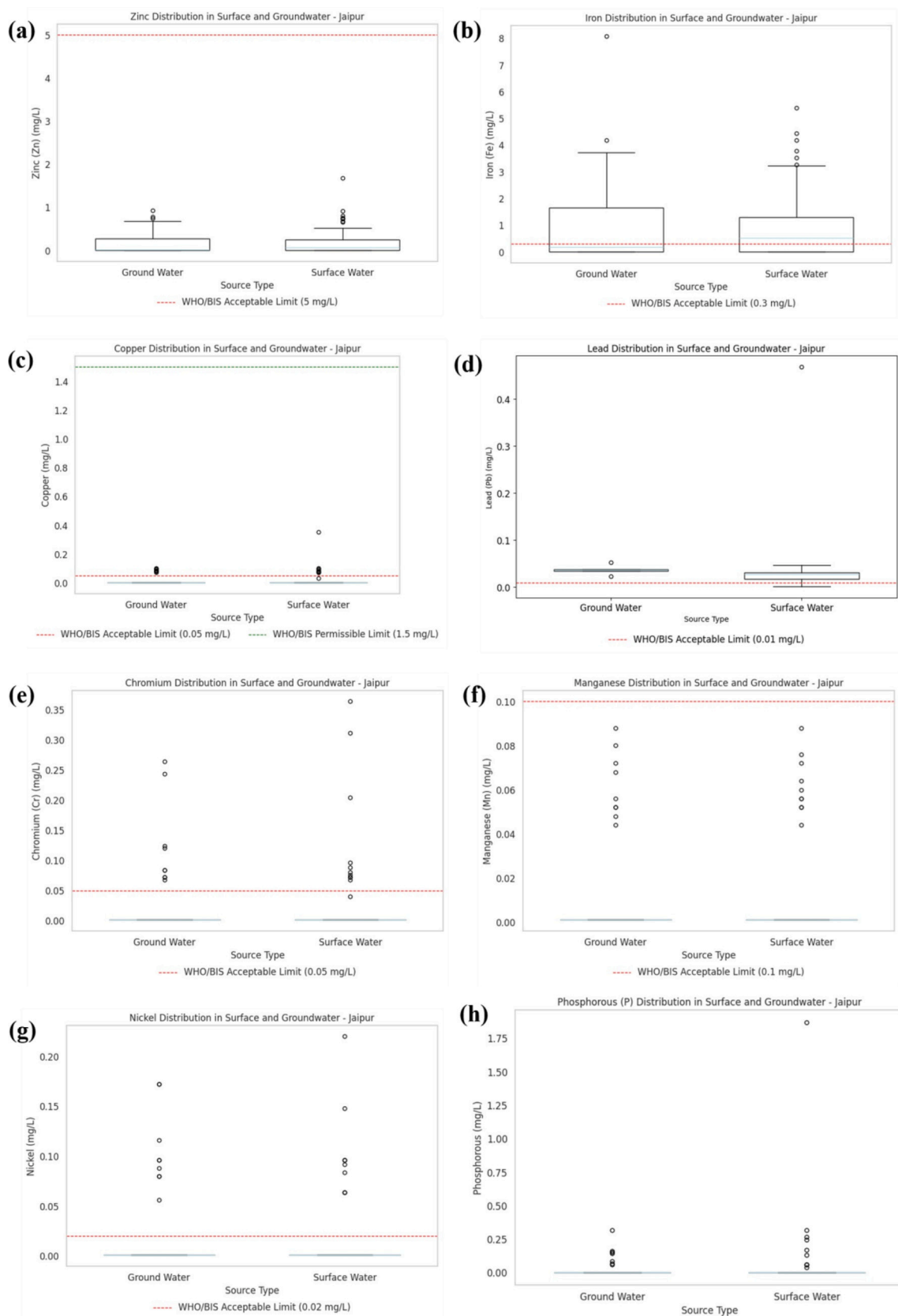


Fig. 4. Box plots showing elemental distribution in surface and groundwater samples.

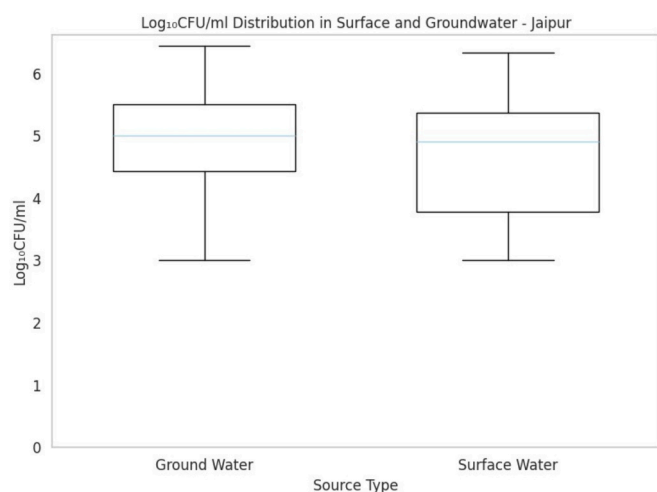


Fig. 5. Box plots showing total aerobic heterotrophs ($\log_{10}$ CFU/mL) distribution in surface and groundwater samples.

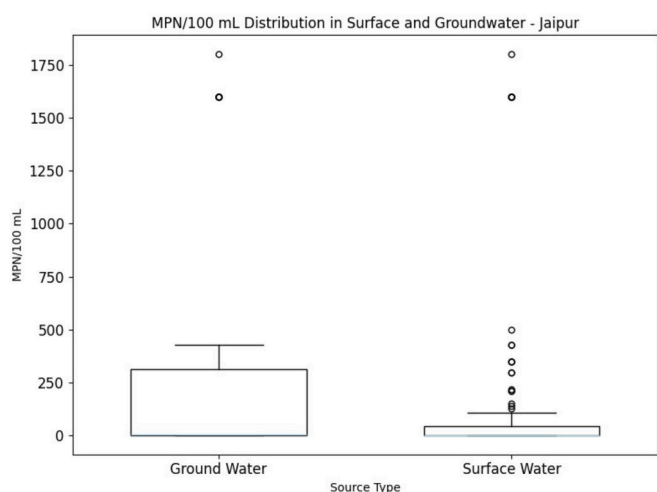


Fig. 6. Box plots showing total coliforms (MPN/100 mL) distribution in surface and groundwater samples.

2024), while *Enterococcus* spp. from groundwater also demonstrated elevated MDR rates (59.1%) (Farkas et al., 2022). The diversity of resistance, spanning 1 to 12 antibiotic classes in the present study, reflects the adaptive responses of these bacteria to environmental pressures, including untreated sewage and persistent antimicrobial residues. These findings are consistent with studies highlighting the role of urban water bodies as hotspots for ARGs dissemination, necessitating urgent surveillance and mitigation strategies to protect public health in semi-arid regions (Junaid et al., 2022; Qing et al., 2024).

MARI values in the present study ranged from 0.2 to 1.0, with 96.45% exceeding the high-risk threshold (MARI >0.2), indicating exposure to antibiotic-contaminated environments (Shahid et al., 2023). The highest MARI (1.0) was observed in groundwater isolate JS98, resistant to 19 of 20 antibiotics tested, underscoring the risk of MDR strains proliferation in polluted settings (Ahmed, 2023). Comparable findings from sewage treatment plant sludge have reported elevated MARI values (up to 0.93) among resistant bacterial isolates, demonstrating the widespread prevalence of AMR in the environment (Patra et al., 2024). The contamination of drinking water sources with MDR bacteria, exacerbated by untreated wastewater discharge, poses severe health risks, reinforcing the urgent need for strengthened sanitation and antibiotic stewardship to curb environmental AMR transmission (Li

et al., 2023).

3.4. Multivariate statistical analysis

3.4.1. Principal component analysis

PCA was applied to reduce dimensionality of the dataset while preserving key variance across physicochemical and microbiological parameters of drinking water samples. The scree plot (Fig. 8) showed a sharp decline in eigenvalues from PC1 (5.48) to PC3 (2.21), together capturing the majority of variance, similar to findings reported in water quality assessments from Algeria (Mohamed Amine et al., 2021), indicating that these components encapsulate critical information about water quality. Components PC4 to PC6 (eigenvalues: 1.83–1.04) contributed secondary variance, revealing latent patterns in water quality data as demonstrated by Tiouiouine et al. (2020). PCA effectively highlighted key drivers of water quality, such as urban runoff and agricultural influence, thereby supporting its use in effective water resource assessment and management (Benkov et al., 2023).

The number of retained components was determined using the Kaiser criterion (eigenvalues >1) and the elbow method, both of which converged at PC6, beyond which additional components offered minimal explanatory value. The first six PCs accounted for 70.78% of total variance, with individual contributions from 26.09% (PC1) to 4.96% (PC6), supporting their selection for further analysis (Sclove, 2022).

The PCA biplot (Fig. 9) illustrated the distribution of samples and variables along PC1 (26.09%) and PC2 (13.85%), together explaining 39.94% of total variance. PC1 was primarily driven by HMs and non-metal – Cr (0.38), Cu (0.37), Mn (0.37), Ni (0.33), P (0.30), alongside nitrate (0.25), indicating a combined geogenic and anthropogenic contamination gradient (Hunter et al., 2024; Zheng et al., 2025). High-loading samples such as J110, J85, J122, J115, and J108, exhibited elevated HMs concentrations, consistent with previous reports linking metal variability to rock–water interactions and localized pollution inputs (Jiang et al., 2023; Nur-E-Alam et al., 2022).

PC2 reflected a microbial-antibiotic resistance gradient, with strong positive loadings for MARI (0.41), *E. coli* (0.37), and total coliforms (0.30), alongside negative loadings for Fe (−0.31), Pb (−0.28), and Zn (−0.28), suggesting an inverse relationship between microbial contamination and trace metals, potentially linked to co-selection mechanisms where metal resistance may influence antibiotic resistance (Vats et al., 2022). Samples like J130 and J145, which align with microbial contamination, contrast with J75 and J70, associated with metal-related contamination, reflecting patterns observed in urban waters studies (Squadrone, 2020).

Vector direction revealed key correlation patterns: variables such as Cr, Cu, and Mn that point in similar directions are positively correlated, while those oriented nearly 180° apart such as $\log_{10}$ CFU/mL and Pb – indicate negative correlations. Groundwater samples (blue) showed greater dispersion along PC1, reflecting greater variability in HMs due to geogenic inputs, whereas surface waters (green) clustered near the center, with exceptions like J110 likely impacted by industrial runoff (Kim et al., 2023; Liu et al., 2021). This spatial distinction underscores the effectiveness of PCA in identifying dominant contamination drivers and associated health risks, consistent with other environmental multivariate studies in complex water quality assessments (Mohammed et al., 2022; Teixeira de Souza et al., 2021).

To further improve interpretability, the six retained components (cumulative variance explained = 70.78%) were subjected to Varimax rotation. Rotation redistributes variance among factors without altering the total explained variance, clarifying underlying data structure and source contributions. The first two rotated components (RCs) accounted for 26.09% (RC1) and 13.85% (RC2) of the variance, together representing 39.94% of total variance. The rotated loadings were visualized on the RC1–RC2 plane (Fig. 10), which clearly illustrates the clustering of variables into distinct contamination factors.

The varimax-rotated structure distinctly revealed two dominant

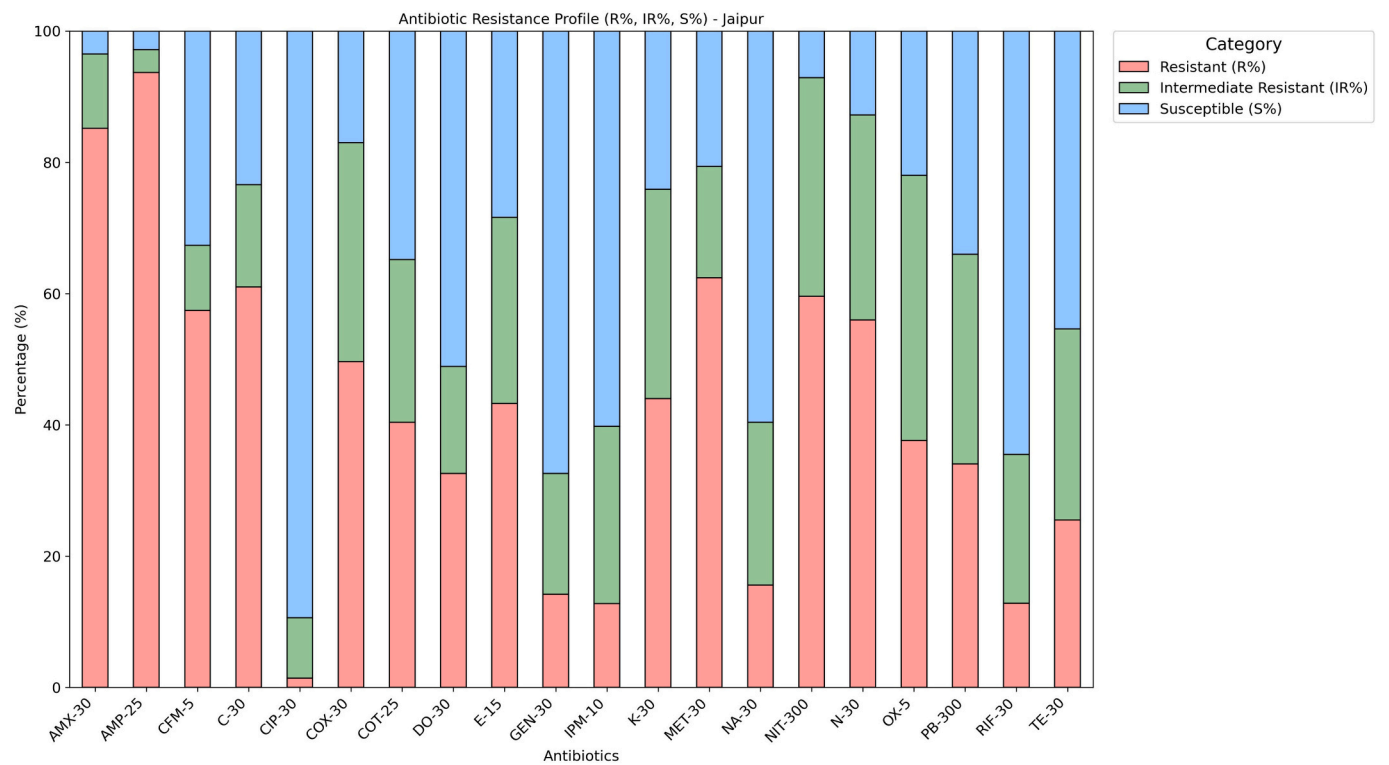


Fig. 7. Stacked bar chart showing the antibiotic resistance profile of 141 *E. coli*-like isolates across multiple antibiotic classes.

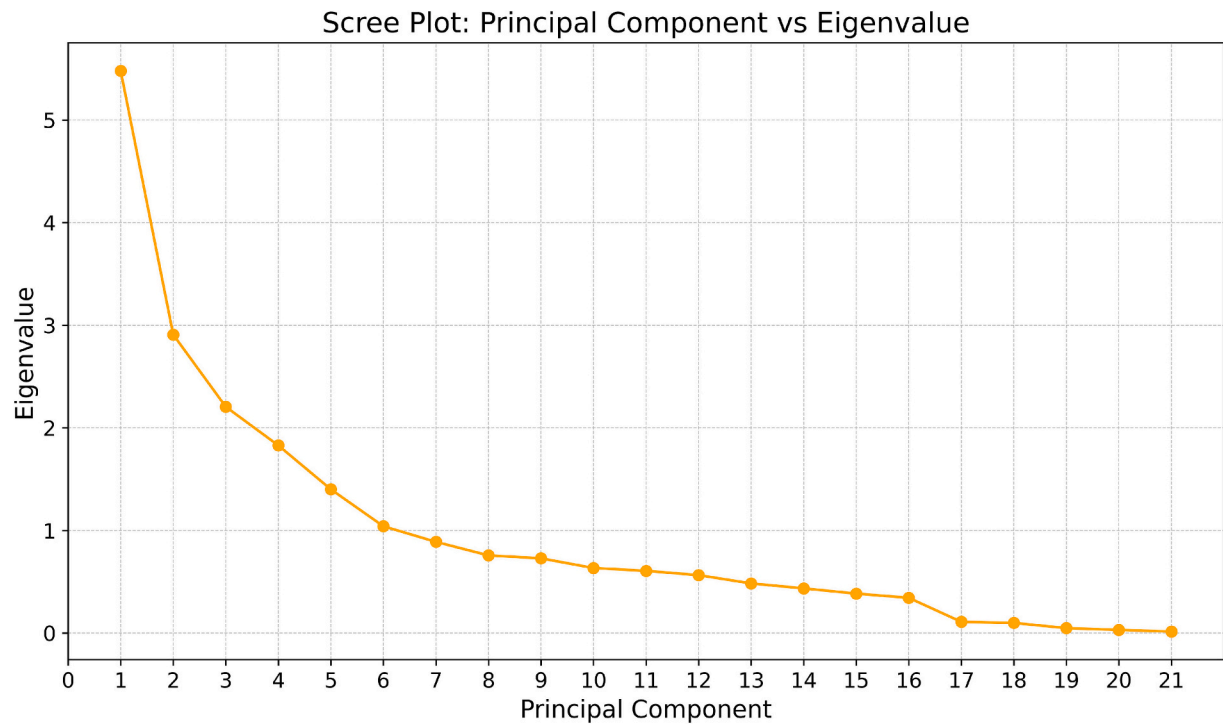


Fig. 8. Scree plot illustrating the eigenvalues corresponding to the first 21 principal components derived from PCA.

contamination modes. RC1 emerged as a metal-dominated factor, with very strong loadings for Ni (0.949), Mn (0.900), and Cr (0.831), along with strong-to-moderate loading from Cu (0.660). This clustering indicates a shared source signature, likely from anthropogenic activities such as industrial effluents and corrosion, compounded by geogenic inputs from aquifer lithology (Faisal et al., 2024). The weaker

associations of Zn (0.285) and P (0.286) with RC1 suggest co-mobilization of nutrients with metals at sites influenced by wastewater discharge. Similar patterns have been observed in the Çanakkale Strait, where Zn and P enrichment was linked to domestic wastewater and agricultural activities (Varol and Tokatlı, 2024).

RC2 was defined as a microbial-AMR-nitrate factor, with extremely

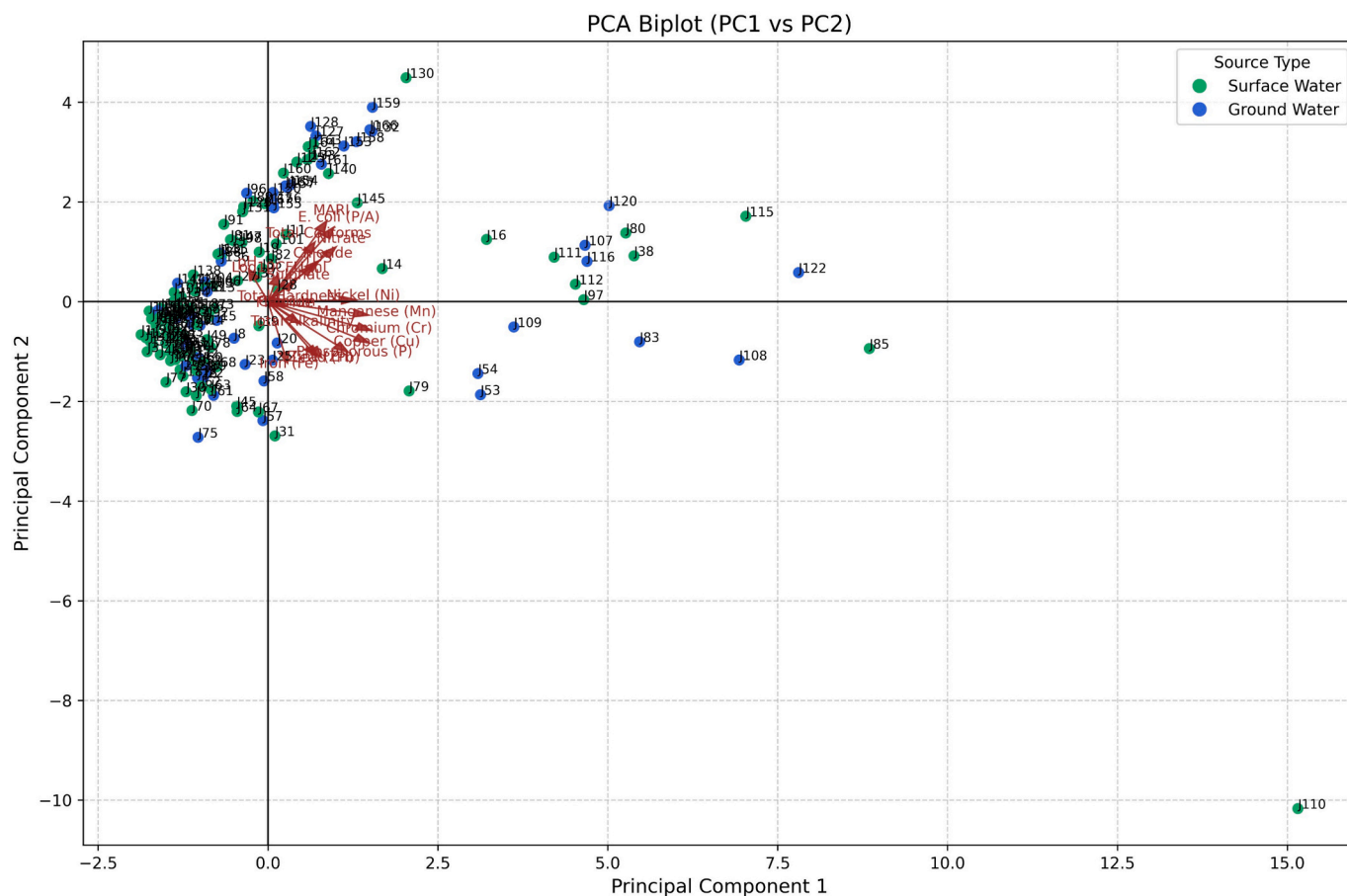


Fig. 9. PCA biplot (PC1 vs. PC2) illustrating sample clusters by water source and key variable loadings indicating major contamination gradients.

high loadings for MARI (0.904) and *E. coli* (0.884), and moderate contributions from total coliforms (0.339) and nitrate (0.309). This grouping highlights fecal contamination and AMR propagation within the study area, with sewage intrusion and agricultural runoff as major contributors (Tarek et al., 2024). The concurrent nitrate loading supports the view that nutrient enrichment often coincides with microbial/AMR pollution (Vadde et al., 2022).

The remaining RCs reflected baseline geochemical processes. RC3 was dominated by total alkalinity (0.934), indicating carbonate buffering and groundwater chemistry influenced by climatic and geological factors such as seawater intrusion and evaporation (Hung et al., 2024). RC4 was defined by TDS (0.931), representing salinization and ionic strength associated with agricultural practices and anthropogenic activities, consistent with findings from karst aquifers affected by fertilizer and runoff inputs (Spellman et al., 2024). These geochemical signals separate natural processes from anthropogenic contamination drivers, emphasizing the value of rotation in source differentiation (Hung et al., 2024). Overall, rotated PCA sharpened the interpretability of parameter groupings, distinctly separating metal contamination from microbial-AMR contamination while also isolating background geochemical influences, which together shape the drinking water quality in the study area. However, DWCI remains based on unrotated PCA scores, ensuring methodological rigor and preserving the novelty of this study.

3.4.2. Drinking water contamination index assessment

The formulated PCA-derived DWCI provided a robust, multivariate framework for quantifying contamination levels across surface and groundwater samples (Fig. 11). Among surface waters, sample J110 recorded the highest DWCI (22.03) indicating more severe composite contamination, primarily due to elevated Cr, Mn, Cu, coliforms, and

MARI levels, followed by J85 (9.73), J115 (8.65), J80 (6.50), and J38 (6.40), suggesting cumulative impacts from urban runoff and wastewater (Zhang et al., 2022). Groundwater samples J122 (8.74), J120 (8.72), J108 (8.41), J83 (6.42) and J116 (5.83), showed moderate-to-high contamination, reflecting geogenic leaching and surface contaminant infiltration (El-Kholy et al., 2022).

The novelty and utility of DWCI approach offers key advantages over traditional water quality indices by (i) leveraging PCA loadings for objective, parameter-free weighting; (ii) translating multivariate extremes into a continuous contamination scale; and (iii) eliminating sign effects by squaring PC scores, capturing only magnitude (Daga et al., 2020; Arcari et al., 2023; Pronk et al., 2021). These features enable the effective identification of contamination hotspots and allow for straightforward application across diverse hydrochemical datasets with minimal computational demand, which are crucial for integrated assessments in varied water systems (Petculescu et al., 2024), and aligning with recent advances in water quality monitoring using big data and machine learning (Uddin et al., 2024).

3.4.3. Source-wise statistical comparison of DWCI

The Mann-Whitney *U* test revealed a significant difference in DWCI between groundwater and surface water samples ($U = 2238$, $p = 0.0124$, $r(b) = 0.24$), with groundwater showing a higher median DWCI (3.00 vs. 2.51) and greater variability (IQR: 2.24 vs. 1.10; Fig. 12). Groundwater, while less susceptible to episodic contamination, faces challenges from hydrocarbons and other difficult-to-degrade compounds, posing long-term risks to ecosystems and human health (Goel, 2021). Surface water, in contrast, displayed greater DWCI variability and extreme outliers like J110, driven by direct anthropogenic inputs such as sewage and runoff (Karunanidhi et al., 2022). These findings underscore the

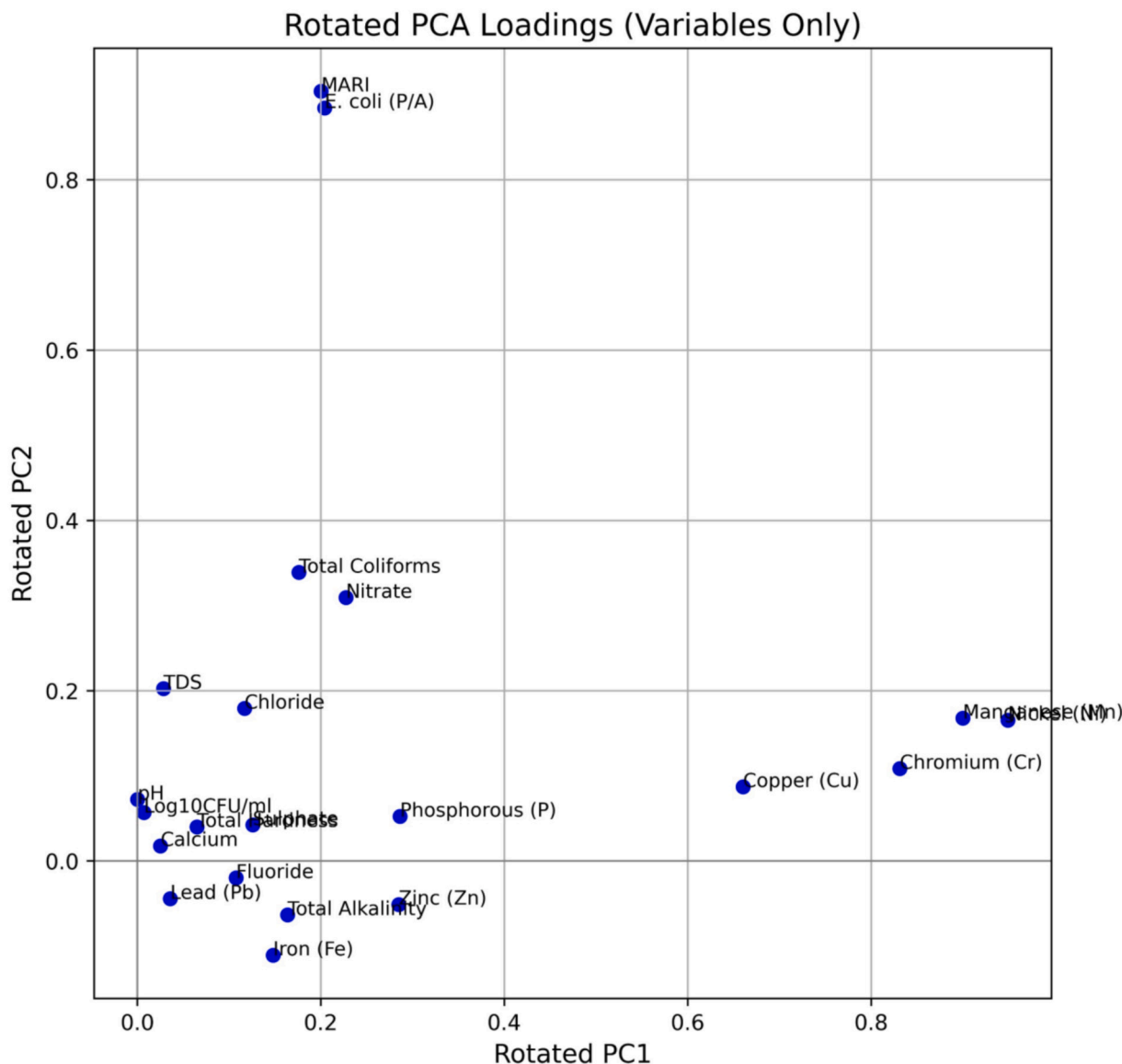


Fig. 10. Rotated PCA loadings (Varimax) plotted on Rotated PC1 vs Rotated PC2.

necessity for dual-source monitoring strategies: real-time surveillance for surface waters and long-term remediation efforts for groundwater to effectively manage both acute and chronic contamination risks.

3.5. Cluster analysis

3.5.1. K-means cluster analysis

The combined F-value and P-value plot (Fig. 13) identified the relative importance and statistical significance of water quality variables in differentiating K-means clusters (Ahmadmoazzam et al., 2021). In this study, F-statistics (bar lengths) indicated the effectiveness of each variable in separating clusters based on the ratio of between- to within-cluster variance, while P-values (violet dots) assess their statistical significance, thereby identifying key drivers of water quality variation (Huang, 2023; Zhang et al., 2022). Consistent with previous research, physicochemical parameters emerged as major contributors to clustering patterns, underscoring the value of KMC for tracking spatial and temporal pollution trends in environmental monitoring and management (Najafi Neysiani et al., 2022; Novianta et al., 2023).

High F-values for P (1828.63), Pb (1166.37), Cu (800.64), Mn (380.57), Ni (374.25) and Cr (273.54), indicated these elements as

primary contributors to water quality variability, largely attributed to industrial discharge and agricultural runoff (Kana, 2022). Statistical analyses showed that variables (HMs) with P-values < 0.05 significantly influenced clustering at the 95% confidence level, while others such as pH, TDS, chloride, nitrate and total alkalinity did not (Liu et al., 2021). Although microbial indicators like MARI (14.01), *E. coli* (13.70), total coliforms (7.44) and Log₁₀CFU/ml (0.68), showed lower statistical influence, they still contributed to group separation, suggesting interactions between metal contamination and microbial resistance (Kachhwaha et al., 2022). These findings highlight key pollutants for prioritization in water treatment and monitoring strategies.

KMC (k = 3), applied to 21 standardized water quality parameters (Fig. 14), identified three distinct contamination profiles, reflecting the inherent complexity of water quality assessment. The largest group, Cluster 1 (n = 149; J1-J15, J17-J37, J39-J52, J55-J79, J81-J82, J84, J86-J96, J98-J106, J113-J114, J117-J119, J121, J123-J167), exhibited low to background contamination, with slightly negative to near zero z-scores across most parameters, including Cu (−0.27), Mn (−0.31), Cr (−0.28), fluoride (−0.07) and microbial indicators such as Log₁₀CFU/ml (−0.03), total coliforms (−0.10), *E. coli* (−0.13), MARI (−0.12). This profile suggests minimal anthropogenic and microbial influence, largely

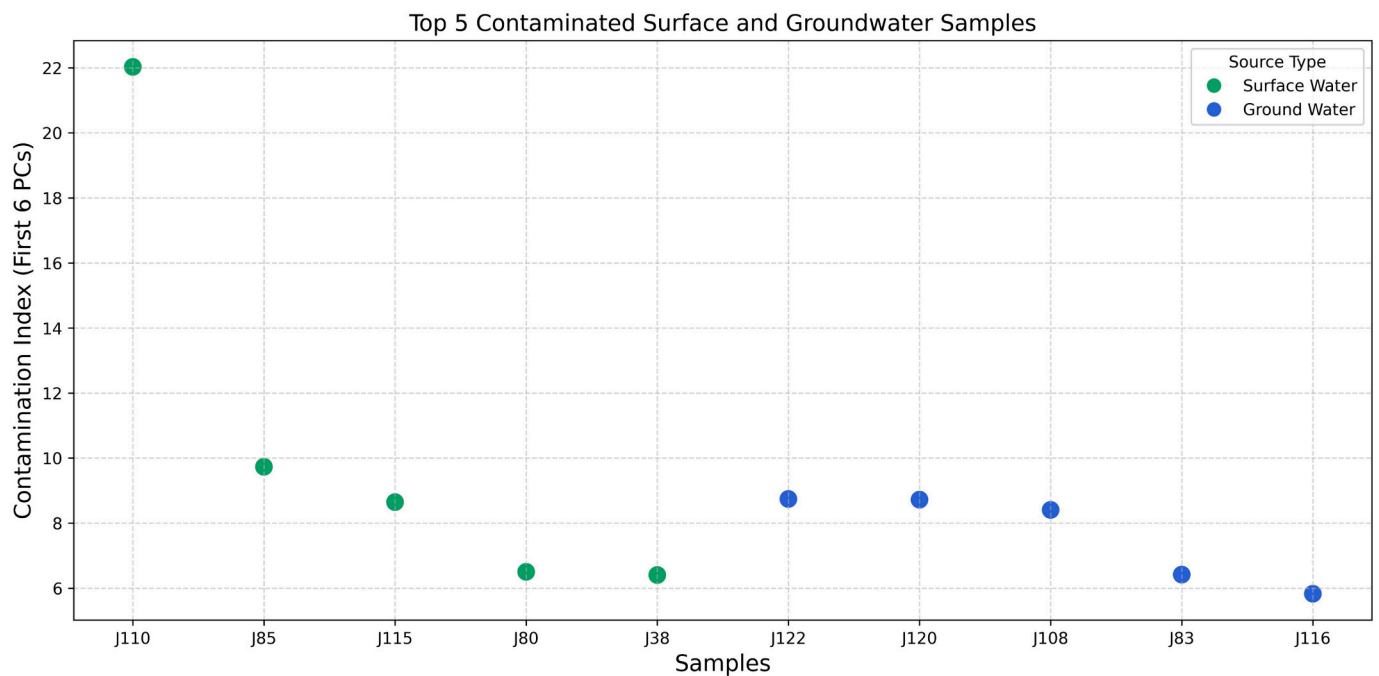


Fig. 11. Top 5 surface and groundwater samples ranked by DWCI using the first six principal components from PCA.

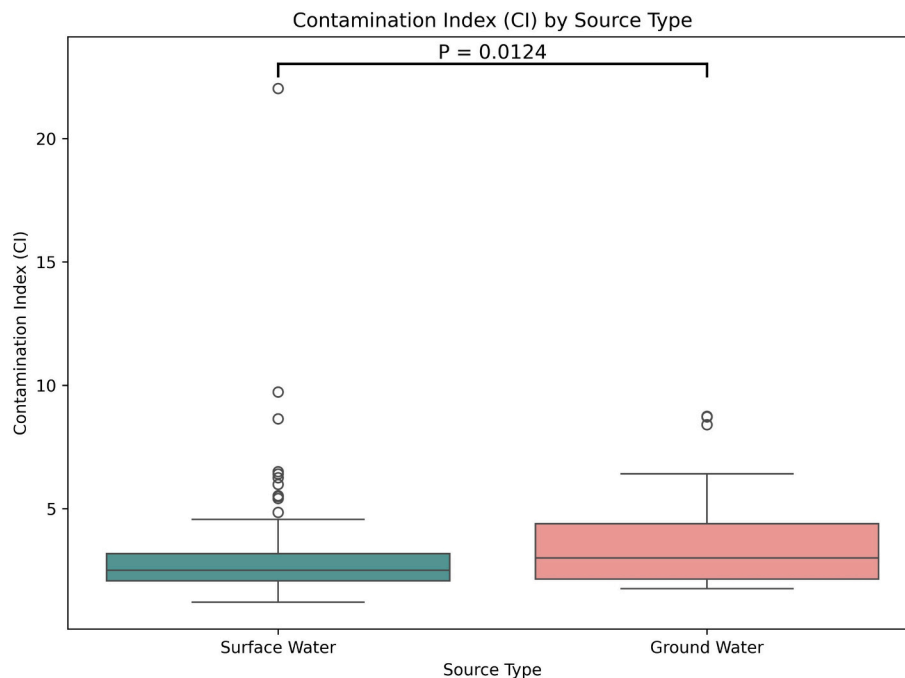


Fig. 12. Statistical comparison of DWCI between surface water and groundwater samples, showing a statistically significant difference (Mann-Whitney U test, $p = 0.0124$).

attributable to natural processes, consistent with observations from protected or low-impact groundwater systems (Reena et al., 2022).

In contrast, Cluster 2 ($n = 17$; J16, J38, J53-J54, J80, J83, J85, J97, J107-J109, J111-J112, J115-J116, J120, J122) exhibited an intermediate contamination profile characterized by elevated levels of metals such as Ni (+2.68), Mn (+2.55), Cr (+2.06), Cu (+1.79), and Zn (+1.19), alongside salinity contributors like nitrate (+1.16), total alkalinity (+0.91), chloride (+0.97) and sulphate (+0.83), suggesting agro-industrial runoff impacts. The presence of *E. coli* (+1.07) and high MARI values (+1.12) indicated moderate fecal and antibiotic resistance

contamination, consistent with patterns in peri-urban environments where co-selection of ARGs and HMRGs is common (Al Salah et al., 2022). Interestingly, chemically contaminated samples showed suppressed microbial indicators including *E. coli* (+1.31), MARI (−0.65), $\log_{10}$ CFU/ml (−0.23), and total coliforms (−0.17), suggesting possible metal-induced inhibition, pointing to localized industrial discharge (Kapoor and Singh, 2021; Kumari and Mishra, 2021).

Cluster 3 ($n = 1$; J110), the smallest but most concerning group, represented an extreme contamination hotspot, marked by sharply elevated levels of Pb (+12.42), P (+12.09), Cu (+9.25), Cr (+6.60), and

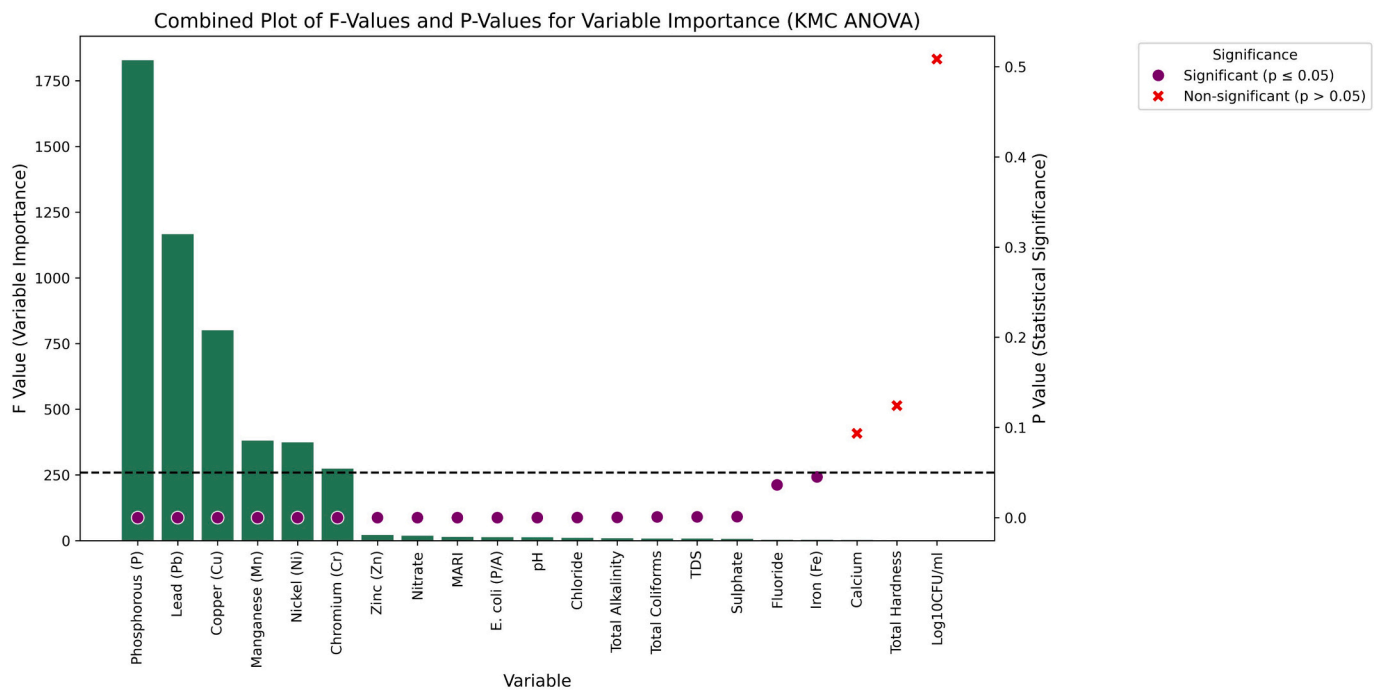


Fig. 13. Combined bar-and-scatter plot of ANOVA F-values (variable importance) and associated P-values (statistical significance) for all 21 variables used in KMC of water samples.

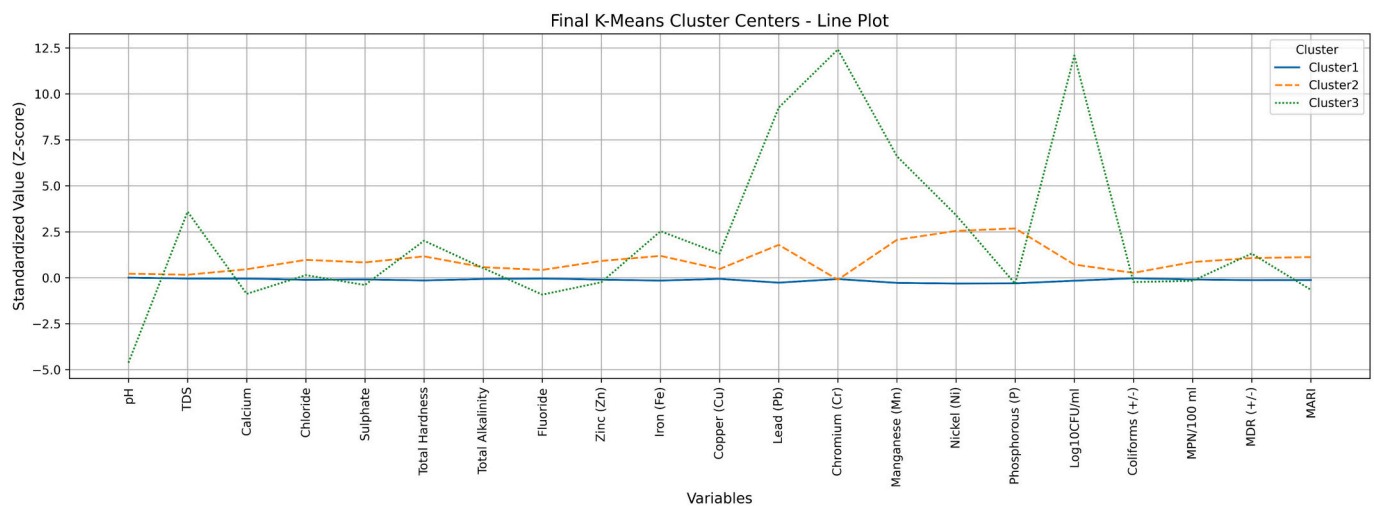


Fig. 14. Standardized K-means cluster-centre profiles ($k = 3$) illustrating contrasting physicochemical-microbiological fingerprints of water samples.

Mn (+3.40), indicating severe anthropogenic pollution similar to that observed in mining-affected regions (Bhatti et al., 2022). Together, these three clusters illustrate a gradient from baseline to intermediate to extreme metal-salinity-microbial contamination, offering a practical framework for source tracking, regulatory prioritization, and risk-informed management of drinking water resources (Nasir et al., 2023).

3.6. Mahalanobis distance analysis for outlier detection

The integration of MD and PCA provides a robust framework for detecting multivariate outliers in water quality assessments (Afzal et al., 2021; Kunjuni and Abraham, 2022). The MD histogram (Fig. 15, left panel) demonstrated a right-skewed distribution, with most values between 2 and 6, indicating a few significant anomalies.

Samples exceeding the 99.5% chi-square threshold ($MD > 6.43$) were flagged as outliers, totaling 22. A detailed list of these samples and

their corresponding MD values is provided in Table 1. Of these 22 outlier samples, 13 were from surface water and 9 from groundwater, indicating that extreme contamination events were more prevalent in surface water, though both sources contributed to high-risk anomalies. These aligned closely with those identified by the PCA-derived DWCI, reinforcing methodological consistency. This finding is consistent with previous studies highlighting MD's value in identifying outliers in non-normally distributed environmental data (Cabana et al., 2021; Teves, 2023). The accompanying scatter plot (Fig. 15, right panel) enhances interpretability by clearly distinguishing normal (green zone) from anomalous (red zone) samples, underscoring the utility of robust multivariate techniques for accurate water quality monitoring and contamination assessment (Bulut, 2020).

The implications of this study underscore the urgent need for integrated water quality management frameworks that connect scientific assessment with environmental policy and public health priorities,

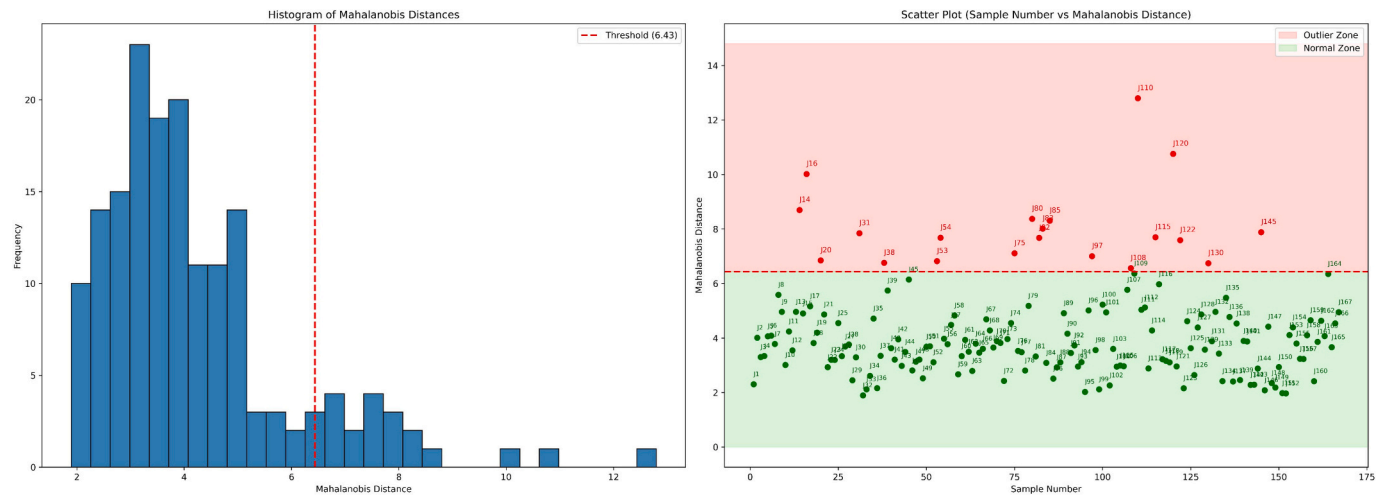


Fig. 15. Histogram of MD (Left) and Scatter Plot of Sample Number vs MD (Right), identifying multivariate outliers beyond the 99.5% confidence threshold (MD > 6.43) in drinking water samples.

Table 1
Outlier water samples identified through MD analysis at the 99.5% chi-square threshold (MD > 6.43).

Source Type	Sample No.	Mahalanobis Distance (MD)
Surface Water	J110	12.80
Ground Water	J120	10.88
Surface Water	J16	10.37
Surface Water	J31	8.86
Surface Water	J14	8.83
Surface Water	J85	8.63
Ground Water	J75	8.62
Surface Water	J80	8.44
Surface Water	J145	8.26
Ground Water	J54	8.21
Surface Water	J82	8.12
Ground Water	J83	8.08
Surface Water	J115	7.93
Ground Water	J20	7.89
Ground Water	J122	7.65
Surface Water	J130	7.62
Surface Water	J97	7.29
Ground Water	J53	7.19
Surface Water	J38	7.16
Ground Water	J108	6.93
Ground Water	J109	6.84
Surface Water	J164	6.51

particularly in semi-arid regions. The co-occurrence of geogenic and anthropogenic contaminants highlights the necessity of coupling routine monitoring with predictive, spatially explicit approaches for proactive intervention. Recent studies have shown that integrating spatial-temporal modeling with multivariate WQIs enhances diagnostic precision and enables predictive mapping of contamination dynamics in groundwater systems (Ustaoglu et al., 2025). Moreover, Taş et al. (2019) emphasized that combining land-use data, pollution source mapping, and real-time decision-support tools can substantially strengthen water management and policy responsiveness in rapidly urbanizing settings. Building on these insights, Tokatli et al. (2024) proposed incorporating spatial and temporal heterogeneity into composite WQIs to improve representativeness across diverse hydrogeological conditions. Further, Karadeniz et al. (2024) demonstrated that groundwater vulnerability mapping integrated with geochemical and health risk models provides actionable insights for prioritizing intervention zones and safeguarding drinking water resources.

In semi-arid regions, these methodological and policy insights suggest that conventional water quality monitoring should evolve to

include continuous microbial and AMR surveillance, spatial mapping of contamination hotspots, and the application of advanced statistical or machine learning models for source identification and predictive assessment. Embedding such approaches into regional policy frameworks would enable data-driven decision-making, targeted aquifer protection, wastewater reuse management, and community-level awareness initiatives. Collectively, these methodologies address the present framework’s limitations, particularly limited spatial integration, static monitoring, and lack of predictive capacity, thereby enhancing the scientific and applied relevance of this study. The findings thus provide a robust baseline for developing region-specific, evidence-based water safety plans aligned with India’s Jal Jeevan Mission and the WHO’s Water Safety Framework.

4. Conclusion

This study provides an integrated assessment of drinking water quality in a semi-arid urban setting, combining chemical, microbial, and antibiotic resistance data within a multivariate framework. Several surface and groundwater samples exceeded WHO limits for TDS, fluoride, Pb, and Cr, while *E. coli* contamination was detected in over one-third of samples, and MDR was widespread, with >96% of isolates exceeding critical MARI thresholds, highlighting the growing environmental burden of AMR. By applying advanced statistical tools, including formulated PCA-derived DWCI, KMC and MD analysis, the study revealed contamination patterns ranging from natural geogenic influence to severe anthropogenic and point-source pollution. Although geographically limited, the applied methodology offers a scalable and replicable framework to other regions facing similar challenges for cumulative contamination assessment. Expanding future work surveillance to incorporate long-term monitoring, molecular AMR profiling, and socio-environmental factors will be key for informing targeted interventions and advancing SDG-6 goals for safe, sustainable drinking water.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jconhyd.2025.104764>.

Availability of data and materials

The experimental data will be provided by the corresponding author upon request.

CRediT authorship contribution statement

Srishti Srivastava: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Abdul Malik:** Writing – review & editing, Supervision.

Consent to participate

All authors consent to participate.

Consent for publication

All authors consent to publish in this journal.

Ethical approval

Not Applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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